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Assessing cross-currency predictability in forex markets: Insights from limit order book data☆

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ABSTRACT

Theoretical research suggests potential cross-currency predictability in currency exchange markets, but empirical findings with macro-based models show mixed results. This has led to a growing focus on micro-based models which explore how trading integrates fundamental information into exchange rates. This study stands out by using detailed limit order book data for the foreign exchange market, enabling micro-level insights. It covers multiple currency pairs, assessing cross-currency predictability. Emphasizing short-term forecasts from one minute to one hour, it introduces factor-augmented regressions, including unsupervised and supervised principal components analysis to tackle data dimensionality, and contrasts those to LASSO and random forest methods. Our findings reveal generally low predictability across various models, supporting the efficient market hypothesis. We find that cross-currency variables offer limited additional insight, but certain microstructure variables like order flow show short-term predictive power, suggesting transient market inefficiencies.

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1. Introduction

Foreign exchange (forex, FX) is by far the largest asset class in the world by volume, with a daily turnover of about $6.6 trillion USD according to the 2019 Triennial Central Bank Survey of Foreign Exchange and OTC Derivatives Markets. In comparison, the estimated average daily turnover in the global stock market is around $84 billion USD per day (BIS, 2019). Despite its size, relatively little research has been done on the detailed functioning (market microstructure) of forex markets compared to equity markets. A similar tendency is observed in the literature regarding exchange rate predictability in contrast to stock return predictability (see e.g. McLean and Pontiff (2016), Reschenhofer et al. (2020) for reviews of research in the latter field).

Exchange rate forecasting has predominantly been considered within a macro-based setup, meaning that the standard models are based on macroeconomic variables, such as interest rates, output, or prices. This strand of literature provides mixed results in terms of exchange rate predictability. For example, Meese and Rogoff (1983) examine various structural models for out-of-sample predictability. They demonstrate that standard macro-based models fail to predict exchange rates. Cheung et al. (2005) examine interest rate parity as well as productivity-based models. They conclude that models that work well in one period do not necessarily work well in another period. Rossi (2013) surveys the existing literature on exchange rate forecasting adopting various econometric methods. She reaches mixed conclusions and points out that they vary depending on a number of choices made

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by the researchers. More recent papers are relatively optimistic in terms of findings. For example, Beckmann et al. (2020) build upon Rossi's work and consider a cross-section of exchange rates to which they apply a modified Bayesian vector autoregressive (VAR) technique allowing for models to vary over time.[1] They find evidence of predictability resulting in economic gains from the cross-section of exchange rates. Dahlquist and Hasseltoft (2020) find long-run (one to six months) predictability based on economic indicators. Hillebrand et al. (2023) find strong predictability, for both sign and return, from factor models with time-varying loadings.

These mixed results in macro-based exchange rate forecasting have spurred interest in so-called micro-based models—for example, those found in Evans and Lyons (2002b, 2005). Such models study how fundamental information is incorporated into exchange rates through trading. The theoretical basis for these models developed by Evans and Lyons (2002b) posits that trades (order flow) may contain information about fundamentals that the market only gradually learns by observing trading behavior. Therefore, short-run predictability based on order flow in FX rates is consistent with their framework (see e.g. Sarno and Taylor (2001) for a survey of literature demonstrating that microstructural elements such as order flow, trading strategies, and information asymmetry play a role in explaining short-term exchange rate dynamics beyond macroeconomic fundamentals). Tyurin (2003), Kozhan and Salmon (2012), and Fletcher and Shawe-Taylor (2013) also use limit order book information to forecast FX rates. Each of the papers considers a single currency pair, for which the authors find evidence of predictability when compared to a simple benchmark model. Kozhan and Salmon (2012) use a larger sample than Tyurin (2003) and Fletcher and Shawe-Taylor (2013) (five weeks, as opposed to one and five days) and find by optimizing a trading algorithm that the order book information led to profitable trading strategies in 2003 but not in 2008. Several other empirical papers, including Rime et al. (2010) as well as Danielsson et al. (2012), find support for predictability. These papers, however, only use order flow and do not incorporate additional information from the limit order book such as order imbalances from posted bid and ask quantities at different levels. One should note, however, that with the rise of algorithmic trading, it became more apparent that a lot of information created by interactions at the exchange may be lost if order flow is used as a single predictor, since the majority of orders transmitted to the exchange are never executed.[2]

Another aspect of the empirical work on micro-based exchange rate models is that most papers only use information from a single currency pair which is the object of interest for the forecaster (see e.g. Kozhan and Salmon (2012)). However, there is likely to exist cross-currency predictability because of both cross-asset learning (Cespa & Foucault, 2014) and cross-asset trading from arbitrageurs (Gromb & Vayanos, 2010). Cespa and Foucault (2014) propose a theoretical model where liquidity providers (market makers or limit order suppliers) learn information about one asset from related assets. Based on this model, the order book information from different currency pairs should be informative about the future price of one currency, especially when the base currency is the same. For example, the USD/YEN order book may contain information that is relevant for the USD/EUR exchange rate. Relatedly, arbitrage activity with limits resulting from financial frictions could also lead to cross-asset return predictability, as suggested by Gromb and Vayanos (2010). Related to frictions, differences in liquidity between currency pairs with the same base currency may result in cross-currency predictability. If there is, for example, a shock that leads to an excess supply of USD, a trader will likely liquidate USD holdings using the most traded currency pair (the EUR/USD) and then move on to the less liquid pairs such as the CAD/USD.[3] In relation to this, the empirical work of Evans and Lyons (2002a) and Rime et al. (2010) should be pointed out, as they use cross-currency information in their analysis. However, Evans and Lyons (2002a) only use the concurrent order flow for the analysis, and do not provide any forecasts. Rime et al. (2010), on the other hand, do not report separate results for same-currency and cross-currency specifications, which does not allow for evaluating the contribution to predictability from cross-currency pairs. Cespa et al. (2022) document significant cross-currency return predictability using volume data from the OTC FX market, showing that low-volume currencies tend to experience return reversals, even across currency pairs. Similarly, Menkhoff et al. (2016) show that currency-specific order flows from financial institutions help forecast returns in multiple currency pairs, reflecting information transmission across currencies through dealer networks. Overall, the strongest predictor found in previous literature is order flow (buyer- minus seller-initiated trading volume). Evans and Rime (2016) argue that due to the opaque and fragmented nature of FX markets, learning fundamentals from trading behavior can be very slow, meaning that order flow can predict FX returns several hours or even days ahead. If this is indeed the reason for the strong and long-lasting predictability of order flow, we would expect to see less predictability in our dataset which is more transparent due to its use of a central limit order book.

The discussion of micro-based models and the utilization of order flow and limit order book information for predicting exchange rates inevitably intersects with the broader debate on the efficient market hypothesis (EMH) in forex markets. According to the EMH, financial markets are 'informationally efficient', with asset prices reflecting all available information. However, the mixed results in exchange rate predictability pose questions about the forex market's efficiency. While some studies, as indicated

[1] In their case, there are 10 currencies which are expressed in terms of the US dollar and are end-of-month exchange rates that enter the model as discrete returns.

[2] The cancel-to-trade ratio was on average 17–19 from 2012Q1 to 2021Q2 on the NYSE, depending on the market capitalization of the stock (SEC, 2021).

[3] We thank an anonymous referee for supplying this argument.

by Rossi (2013)'s review, suggest some level of market efficiency due to the challenges in predicting exchange rates using macroeconomic variables, other research, including Evans and Lyons (2002a) and Levich and Potì (2015), indicates potential inefficiencies through microstructure variables. These apparent inefficiencies could be due to the opaque and fragmented nature of forex markets and the slow integration of information into prices via trading activities.

Our contribution to the literature is twofold: First, we use detailed limit order book (LOB) information for the FX market, which allows us to account for micro-type information with respect to currency exchange rates. Second, we consider several currency pairs, which enables us to examine the presence of cross-currency predictability. Our focus is on short-run forecasts based on the FX market-microstructure literature in the spirit of Evans and Lyons (2005).

We use LOB information on five currency pairs from the LMAX exchange in London, a platform that has not been previously considered in the literature. In addition, we check the robustness of our results based on the data from the EBS exchange.[4] We construct a set of predictors at several frequencies, which we discuss in detail in Section 2. The raw dataset serves as a source for a large set of potentially highly correlated predictors. In such a case, using the entire set of variables directly in predictive regression models will likely lead to unsatisfactory outcomes, due to multicollinearity, over-fitting, reduced interpretability, and increased computational complexity. To address these challenges, we apply dimension reduction techniques, starting with factor-augmented regression, as per Stock and Watson (2002). This involves identifying common factors from a large set of predictors for use in predictive models. Factor-augmented regressions have shown efficacy in fields like macroeconomics (Stock & Watson, 2016) and asset pricing (Giglio & Xiu, 2021), particularly for forecasting stock returns, as demonstrated by Giovannelli et al. (2021) and Lima and Godeiro (2023). However, its application in FX market forecasting using LOB data is less explored, a gap we aim to fill.

Principal component analysis (PCA), commonly used for factor estimation, has its limitations. We therefore consider supervised principal component analysis (S-PCA) (Bair et al. (2006)) for improved accuracy. Our findings suggest that S-PCA enhances forecast quality. We also compare these models to LASSO (Tibshirani (1996)) and its adaptive version (Zou (2006)), along with random forests (RF) (Breiman (2001)), using a historical-average model as a benchmark. Details are further elaborated in Section 3.

In Section 4, we investigate a variety of cases, each involving different subsets of predictors. Using a rolling window approach, we adapt to potential shifts in market behavior and generate out-of-sample forecasts, considering a broad range of window sizes. Our findings reveal a generally low predictability in exchange rates across all cases and methods, with most models underperforming compared to the historical average benchmark. This supports the efficient market hypothesis, as it suggests that forex markets are effective at integrating available information. Interestingly, we observed that cross-currency variables contribute little to the predictive models, underscoring the rapid integration of information across different currency pairs, a further testament to market efficiency.

Notably, when focusing solely on predictors from the same currency, particularly using PCA, we see a notable improvement in forecast performance for all currency pairs. This contrasts with methods like S-PCA, LASSO, and RF, which show less sensitivity to the inclusion of potentially irrelevant predictors. Despite the overall limited predictability, we identify instances where certain variables, such as order flow, demonstrate predictive power for specific periods. This is particularly evident in the short term, where order flow emerges as a significant predictor for one-minute-frequency returns using the LASSO method. These instances of short-term predictive power, especially in high-frequency data, hint at transient market inefficiencies where the market may not immediately reflect all information. However, across different forecast frequencies, the predictability remains generally low, aligning with the broader implications of market efficiency in forex trading.

2. Data

Forex markets have traditionally been over the counter (OTC) and are therefore opaque to both participants and researchers. However, since the introduction of the Reuters platform and the Electronic Broking Services (EBS) in the late 1990s, it is now possible to investigate the functioning of forex markets using the rich information contained in electronic limit order books (LOBs).

Our main dataset consists of the LOB information from the LMAX exchange in London from October 1, 2019, to April 19, 2020, and we complement this with a dataset from EBS (see Appendix B for details). LMAX is a trading platform used by e.g. banks, funds, proprietary trading firms, brokerages, and asset managers. LMAX has about 5% of the total FX trading volume[5] and is open around the clock. An order book is the collection of bid and ask prices and quantities submitted to the exchange. The highest (best) bid and the lowest (best) ask prices are called 'level one' of the order book, the second-best price is called 'level two', and so on. In turn, a limit order book is a record of outstanding limit orders, which are orders to buy or sell a security at a specific price or better. The data at hand undergo an update every time there is any change in the LOB, which typically happens several hundred times per

[4] Fletcher and Shawe-Taylor (2013) also use EBS exchange data for their analysis. Meanwhile, Evans and Lyons (2002b), Tyurin (2003), Rime et al. (2010), Kozhan and Salmon (2012), and Danielsson et al. (2012) rely on the interdealer transactions from Reuters' trading system. Only Evans and Lyons (2005) use a different data source for the order book. In particular, they analyze all of Citibank's end-user trades.

[5] (https://www.financemagnates.com/institutional-forex/execution/fxspotstream-cboe-and-lmax-exchange-gain-efx-market-share/)

Table 1
Example of the data entries at hand.

Time	Symbol	Bid1	BidQ1	Bid2	BidQ2	Ask1	AskQ1	Ask2	AskQ2
2019-11-01 00:05:00.021	GBPUSD	1.2936	50	1.2936	50	1.2937	50	1.2938	50
2019-11-01 00:05:00.138	AUDUSD	0.6889	200	0.6889	150	0.6889	100	0.6889	400
2019-11-01 00:05:00.385	USDCAD	1.3171	250	1.3170	100	1.3171	50	1.3172	150
2019-11-01 00:05:01.081	USDJPY	107.9890	200	107.9870	275	107.9920	50	107.9930	50
2019-11-01 00:05:03.847	EURUSD	1.1153	125	1.1153	50	1.1153	39	1.1153	100

Note: 'Time' gives the timestamp with millisecond precision. 'Symbol' is the currency pair. 'Bid1' is the best bid price, and 'BidQ1' is the quantity someone is willing to buy at this price. 'Bid2' is the second-best bid price, with 'Ask' being equivalently defined for the sell prices and quantities. The quantities are given in terms of the base currency quoted in thousands.

Table 2
Order book information.

Notation	Description
P_i^{bid}, P_i^{ask}	Best bid and ask prices, where i is the currency pair
P_{ij}^{bid}, P_{ij}^{ask}	Bid and ask prices at different levels of the order book, $j = 2, \ldots, 5$, where P_{i2}^{bid} is the second-best bid price, and so on.
Q_i^{bid}, Q_i^{ask}	Best bid and ask quotes for each currency pair, i.
Q_{ij}^{bid}, Q_{ij}^{ask}	Bid and ask quotes (volume) at different levels of the order book, $j = 2, \ldots, 5$, where Q_{i2}^{bid} is the second-best bid quote for each of the currency pairs, and so on.

Note: We look at all the currencies in relation to the US dollar, so i = USD/CAD, AUD/USD, GBP/USD, USD/JPY, and EUR/USD.

second.[6] The initially available data consist of the 10 best levels in the order book for each currency pair. However, due to the incompleteness of the observations on higher levels, we only look at the five best levels. Table 1 shows an example of the data for the first two levels of the order book.

One should note that the original information available in the LOB is on a tick-by-tick basis, which is typically unequally spaced in time and noisy and therefore difficult to analyze. Given the interest in the underlying common components in the behavior of the currency pairs, we therefore consider the data at different frequencies, ranging from one minute to one hour. We choose to look at the currency pairs in relation to the US dollar. In particular, we consider the following pairs: US dollar – Canadian dollar (USD/CAD), Australian dollar – US dollar (AUD/USD), Pound sterling – US dollar (GBP/USD), US dollar – Japanese yen (USD/JPY), and euro – US dollar (EUR/USD). The choice comes from the interest in cross-currency pair predictability, as we expect the effect to be stronger in cases where a common currency is present.

Based on the LOB information, the bid and ask prices (quotes) and volumes (quantities) are available directly for each level of the order book. Table 2 provides the list of those variables.

Using this information, we construct a set of variables. We begin with obtaining the variable of interest, which is the log return[7] calculated for each of the currency pairs. It is based on the mid-quote (mid-price), which is a common proxy for the transaction price when the latter is not available. In particular, let P_i^{bid} and P_i^{ask} be the best bid and ask prices, respectively, where i is the currency pair. The mid-quote is then obtained as $P_t^M = 0.5(P_t^{bid} + P_t^{ask})$, which in turn gives the variable of interest $y_t = ln(P_t^M / P_{t-1}^M)$, which is the named log price change. Next, using the order book information, we construct a large set of predictors, where we base our choice to a large extent on (Kozhan & Salmon, 2012). Table 3 provides the list of constructed predictors as well as the target variable.

In terms of predictors, the best-known (micro-based) predictor is probably order flow. It is measured as the difference between the transaction quantity (volume) initiated by the buyer and by the seller, respectively. The transaction quantity is calculated every time there is a decrease in the bid or ask quantity in the order book. If the bid quantity decreases, the order is seller-initiated, and if the ask quantity decreases, the order is buyer-initiated. In their early work, Evans and Lyons (2002b) show that the order flow explains changes in FX rates remarkably well. Note however that Evans and Lyons (2002b) regress concurrent order flow on FX returns, so it is not a predictive regression. Evans and Lyons (2005), on the other hand, run predictive regressions and find some predictability, surprisingly mostly at longer forecast horizons (10 to 20 days).

To capture short-term buy and sell demand separately, we use the order book imbalance, defined as the ask quantity over the bid quantity, for the five first levels of the order book as well as the slope of the bid and ask curve. We include the changes in the bid and ask prices, which should capture potential autocorrelation in the price changes. Meanwhile, the log of change in volumes on both the bid and ask sides could reflect traders' reactions to both publicly available information on the LOB and private information from customer orders. Finally, we include the spread, the difference between the bid and ask prices, as it is linked to illiquidity in the

[6] We obtain the tick-by-tick data through a trading company, ValidAlpha.

[7] As pointed out by one of the referees, some studies, for example Ca'Zorzi and Rubaszek (2020), find predictability in levels rather than returns. Their results, however, rely on monthly data and long-run reversion. At the higher frequencies considered here, dynamics are likely dominated by microstructure effects, so predictability, if present, appears in returns rather than levels.

Table 3
Predictors based on the order book information per currency pair.

Notation	Description
P_t^M	Mid-quote: $0.5(P_t^{bid} + P_t^{ask})$
y_t	Log price change: $\ln(P_t^M / P_{t-1}^M)$
$OrdFlow_t$	Difference between the transaction quantity initiated by the buyer and by the seller.
$Imb_{best,t}$	Best ask to best bid volume ratio: Q_t^{ask}/Q_t^{bid}
Imb_{jt}	j-level ask to bid volume ratio: $Q_{jt}^{ask}/Q_{jt}^{bid}$, $j = 2, \ldots, 5$
$Slope_t^{bid}$, $Slope_t^{ask}$	$(P_t^{bid} - P_{2t}^{bid})/Q_t^{bid}$, $(P_t^{ask} - P_{2t}^{ask})/Q_t^{ask}$
$\Delta \ln(Q_t^{bid})$, $\Delta \ln(Q_t^{ask})$	Change in ln of the market depth quoted on the bid and ask sides, i.e. the value of currency (expressed in mln. dollars) available at the best bid, ask prices. $= \begin{cases} \ln(Q_t^{bid}) - \ln(Q_{t-1}^{bid}), & \text{if } \Delta P_t^{bid} = 0 \\ 0, & \text{otherwise} \end{cases}$
ΔP_t^{bid}, ΔP_t^{ask}	Change of bid and ask prices between last and second-to-last events: $P_t^{bid} - P_{t-1}^{bid}$
$Spread_t$	Size of the bid–ask price spread when it is positive: $max(0, P_t^{ask} - P_t^{bid})$
AWS_t	Average weighted spread, as in Ye and Florescu (2019).

Note: The currency pairs of interest: USD/CAD, AUD/USD, GBP/USD, USD/JPY, and EUR/USD. t denotes the current time period, while $t-1$ denotes the previous time period.

Table 4
Descriptive statistics of currency returns.

Currency	Average # orders	# obs.	Daily volume	Q1	Mean	Q3	St. dev.
One minute							
USD/CAD	918	201,037	821	−0.00377	0.00002	0.00378	0.01402
AUD/USD	839	201,037	1460	−0.00643	−0.00002	0.00606	0.02600
GBP/USD	1352	201,037	876	−0.00615	0.00000	0.00610	0.02036
USD/JPY	2255	201,037	650	−0.00458	0.00002	0.00458	0.01570
EUR/USD	2005	201,037	1345	−0.00406	−0.00001	0.00407	0.01375
One hour							
USD/CAD	55,075	4840	821	−0.01604	0.00085	0.01588	0.09308
AUD/USD	50,353	4840	1460	−0.02732	−0.00100	0.02729	0.16962
GBP/USD	81,136	4840	876	−0.02338	0.00039	0.02372	0.13119
USD/JPY	135,272	4840	650	−0.01522	0.00063	0.01609	0.09778
EUR/USD	120,320	4840	1345	−0.01614	−0.00054	0.01477	0.08221

Note: Descriptive statistics are for percentage returns at the one-minute and one-hour frequencies. 'Average # orders' is the average number of updates in the order book each minute and each hour, respectively. '# obs.' is the number of time series observations. 'Daily volume' is the average daily volume in millions of USD. 'Q1' and 'Q3' are the first and third quartiles. 'St. dev.' is the standard deviation.

market microstructure literature and is expected to influence the types of orders submitted. We also include the activity-weighted spread proposed by Ye and Florescu (2019), which is a spread measure that takes into account the trading activity in the interval between when the spread is calculated.

From Table 4, we see that the average number of orders entering the LOB every minute is around 1000 for USD/CAD, AUD/CAD, and GBP/USD, and about twice that for USD/JPY and EUR/USD. The dataset consists of 201,037 observations at the one-minute frequency and 4840 at the hourly frequency.[8] Both the mean and the median return are very close to zero, as expected for high-frequency returns. The standard deviation is highest for the AUD/USD currency pair and lowest for EUR/USD. The standard deviation for the hourly returns is lower than $\sqrt{60}$ times the one-minute standard deviation for all currency pairs, showing that returns are not independently and identically distributed. Comparing the first and third quartile of returns shows that the returns have very little skewness.

3. Methodology

3.1. Predictive regression

Let y_t be the time series variable to be forecasted, and X_t be a set of candidate predictors, where $t = 1, \ldots, T$. In case of interest in obtaining a one-step-ahead prediction, the following equation is estimated:

$$y_{t+1} = \beta_X' X_t + \beta_W' W_t + \varepsilon_{t+1}, \quad (1)$$

where y_{t+1} is a scalar, X_t is an N-dimensional vector of predictor variables, W_t is a vector of predetermined variables such as a constant or lags of y_t, and ε_{t+1} is an idiosyncratic error.

This setup is common in the forecasting context and is referred to as a predictive regression. However, in case of a large-scale dataset (i.e. a large number of candidate predictors N), the estimation of (1) faces a number of challenges, such as multicollinearity, over-fitting, and model complexity. In order to tackle those issues, further modifications of the predictive regression approach are required. Factor models offer a possible solution that

[8] The reason the one-minute frequency has fewer than 60 times the observations of the hourly frequency is that missing observations are more common at the one-minute frequency. The reason all currency pairs have the same number of observations is that if at least one currency pair has a missing observation, that observation is removed for all currency pairs.

allows us to summarize and extract information from large-scale datasets; see Fan et al. (2021) for a review of recent developments in this direction. An alternative solution is to use variable selection and/or shrinkage methods, examples of which are the least absolute shrinkage and selection operator (LASSO) of Tibshirani (1996) or the random forests (RF) of Breiman (2001).

3.2. Factor-augmented regressions

In order to tackle the issue of a large number of predictors, one can assume that the information in this set can be summarized by a smaller number of variables (factors). In particular,

$$X_t = \Lambda F_t + e_t, \tag{2}$$

where F_t is an unobserved $r \times 1$ vector of common factors, Λ is an $N \times r$ matrix of factor loadings, and e_t is the idiosyncratic error. Given that $r < N$, one can then use the common factors instead of the actual predictors in the regression, avoiding the issues discussed above. Then, the predictive regression of interest is

$$y_{t+1} = \beta_F' F_t + \beta_W' W_t + \varepsilon_{t+1}. \tag{3}$$

However, the factors are not observed, so there is a need for their estimation. In order to obtain the forecast of the target variable, $\hat{y}_{t+1}$, a two-step procedure is usually implemented. In the first step, the estimates of the unobserved factors are obtained. In the second, the predictive regression under consideration is run on the estimates of the factors. In this case, a natural question arises of how to obtain the factor estimates.

Stock and Watson (2002) suggest using principal components analysis (PCA) for factor estimation, where factor estimates are obtained as a solution to the following minimization problem:

$$\min_{F_1,\ldots,F_T,\Lambda} (NT)^{-1} \sum_{t=1}^{T} \sum_{i=1}^{N} (x_{it} - \lambda_i' F_t)^2, \tag{4}$$

subject to $\Lambda'\Lambda = I_r$, where λ_i is a $1 \times r$ vector of factor loadings.

We denote by $\hat{F}_t$ and $\hat{\Lambda}$ the minimizers of (4). The solution to the minimization problem is then

$$\hat{F}_t = X_t' \hat{\Lambda}/N, \tag{5}$$

where $\hat{\Lambda}$ corresponds to the matrix of r eigenvectors of the sample covariance matrix $\hat{\Sigma}_X = T^{-1} \sum_{t=1}^{T} X_t X_t'$ associated with its r largest eigenvalues. We can then use those estimates in (3) to obtain the forecasts of the target variable. In practice, one also needs to determine r, the number of principal components to use. There is a whole strand of literature regarding the estimation of the number of factors. For example, Stock and Watson (1998), Bai and Ng (2002), Alessi et al. (2010), and Djogbenou (2021) suggest using information criteria in order to obtain a consistent estimate of r. However, those approaches tend to exhibit limited performance in finite samples. In order to reduce the estimation error, we choose to skip the step of determining the number of factors to use in the factor-augmented regression. Instead, we consider a set of values for r and look at the performance of models with different numbers of underlying latent components.

Even though PCA is widely used for estimating common factors in factor-augmented regressions, it comes with a limitation in the context of predictive regressions. In particular, it does not take into account the relation of the predictors to the variable of interest, as pointed out by Bai and Ng (2008). They note that in the initial implementation, the extraction of factors from the large dataset is independent of the series to be forecasted. So even though the estimated factors might well capture the variation in X_t, it is not guaranteed that they will be informative for predicting y_t. It can instead be the case that there is a subset of the original large dataset that can give better results in the forecasting exercise. There are several ways of dealing with this issue which focus on selecting a subset of predictors. For example, Bai and Ng (2008) refer to the question of using pre-selected predictors for forecasting purposes as 'targeted predictors', where the selection can take place using hard and soft thresholding rules, which we consider further.

3.3. Hard thresholding

The method of hard thresholding exploits a particular rule to determine if the ith predictor is significant for predicting the outcome variable without regard for the other predictors being considered. Bair et al. (2006), suggest applying hard thresholding in the context of PCA. They suggest a supervised version of PCA (S-PCA). Though originating in genetic studies, their pre-selection technique has found its way to economics; see for example Jurado et al. (2015).

To take into account the relation of the predictors to the target variable, Bair et al. (2006) suggest the following:

1. Estimate coefficients from running a univariate regression of the target variable on each of the predictors separately.
2. Form a reduced data matrix consisting of only those predictors whose univariate coefficient exceeds a threshold θ in absolute value (θ is estimated by cross-validation).
3. Compute the first (or first few) principal components of the reduced data matrix.
4. Use the(se) principal component(s) in the predictive regression framework.

This procedure is similar to that of Stock and Watson (2002), where the main difference lies in the selection step that takes place prior to estimating the common factors. This allows for controlling for the strength of the relation between the predictors and the target variable.

3.4. Soft thresholding

The method of soft thresholding performs subset selection and shrinkage simultaneously, in contrast to PCA, which uses all of the information to construct the factor estimates. We focus on one such method, namely the LASSO estimation of Tibshirani (1996). It allows for

dropping uninformative predictors using a penalized regression prior to the forecasting exercise, by solving the following minimization problem:

$$\min_{\beta_X}\left[\sum_{t=1}^{T-1}(y_{t+1}-\beta_0-\beta_X' X_t)^2+\delta\sum_{i=1}^{N}|\beta_{X,i}|\right], \tag{6}$$

where the second term is an l_1 penalty function that depends on the penalty parameter δ. This penalty term allows LASSO to shrink irrelevant variables to zero, therefore performing selection. LASSO has been widely used for stock return prediction (see e.g. Zhang et al. (2020) and the references therein).

The shrinkage literature considers various alternative penalties, for example ridge regression and elastic net. In contrast to LASSO, these penalty functions perform shrinkage but not selection. Given that we have a relatively large set of potential predictors, where we suspect that the majority potentially have low or no predictive power, we prefer using the LASSO to other penalties in order to be able to select the predictors. We choose the degree of regularization by using five-fold cross-validation. Given that there is no autoregressive component in the LASSO representation in question, regular cross-validation applies, as the relationship between the outcome variable and predictors is preserved in every draw.

LASSO performs variable selection by shrinking the coefficients of less important predictors towards zero, effectively excluding them from the model. However, when predictors are highly correlated, LASSO tends to arbitrarily select only one predictor among the correlated group while shrinking the others to zero. The specific predictor selected can be sensitive to small changes in the data, leading to instability and inconsistency in the selected model.

To mitigate the inconsistency problem of LASSO in the presence of highly correlated predictors, Zou (2006) proposes a modification to the approach, the adaptive LASSO. He addresses the issue of correlated predictors by using adaptive weights for penalizing different coefficients in the l_1 penalty:

$$\min_{\beta_X}\left[\sum_{t=1}^{T-1}(y_{t+1}-\beta_0-\beta_X' X_t)^2+\delta\sum_{i=1}^{n}\omega_i|\beta_{X,i}|\right]. \tag{7}$$

The penalty adjustment/weights are $\omega_i = |\beta^*_{X,i}|^{-1}$. For $\beta^*_{X,i}$, Zou (2006) suggests using $\hat{\beta}_{ols}$ from an OLS regression.

3.5. Random forests

So far, we have assumed the linearity of our model. However, we would also like to investigate non-linear predictability, for which we consider the following model:

$$y_{t+1} = f(X_t) + u_{t+1}, \tag{8}$$

where f is some function, and u is the idiosyncratic error. There are several non-linear machine learning approaches that can be pursued from this point onward. We focus on tree-based models, as they accommodate complex interactions and other non-linearities among the predictors. Out of the tree-based methods, the random forests (RF) of Breiman (2001) is the preferred one, as it produces trees that are less correlated with each other. An RF is a collection of regression trees, each specified in a bootstrap sample of the original data. In turn, a regression tree is a non-parametric model that approximates the unknown non-linear function $f(X_t)$ in (8) with local predictions using recursive partitioning of the space of the covariates.

Given the predictors $X_t = (x_{1,t}, x_{2,t}, x_{N,t})'$, all possible values of X_t are split into J distinct and non-overlapping regions $S_1, \ldots, S_J$. Then, the regression tree method provides a constant estimate within each region. In particular, it approximates $f(X_t)$ by a function g, such that

$$g(X_t) = \sum_{j=1}^{J}\hat{c}_j 1_{X_t \in S_j}, \tag{9}$$

where $\hat{c}_j$ is the average of y_{t+1} for the subsample with $X_t \in S_j$. The J regions are constructed by sequentially splitting the space into high-dimensional rectangles through recursive binary splitting:

$$S_1(t,s) = \{X|X_t \le s\} \text{ and } S_2(t,s) = \{X|X_t > s\}, \tag{10}$$

where i and s are chosen to minimize the sum of squared residuals

$$\sum_{j=1}^{J}\sum_{t\in S_j}(y_t - \hat{y}_{S_j})^2, \tag{11}$$

where $\hat{y}_{S_j}$ is the mean response for the training observations within the jth box.

Recall that an RF is based on a number of decision trees on bootstrapped training samples. Therefore, the final forecast is the average of the forecasts of regression trees for each of the samples. When building those trees, each time a split in a tree is considered, a random sample of m predictors is chosen from the full set of N predictors. Only those m predictors are then used at the particular split. For the empirical analysis, we set the choice set of predictors at each node to $m = \sqrt{N}$, as commonly done in the literature.

3.6. Estimation and comparison

Our estimation for all models under consideration relies on a rolling window strategy; that is, we use a rolling window of size R to produce the forecast. We use a pseudo-out-of-sample (OOS) approach to evaluate the performance of the forecasts by different models. This allows us to avoid the look-ahead bias and adapt the forecasting process for future use. We consider several lengths of the rolling window in the empirical analysis.

All estimation, including factor selection, as well as machine learning and shrinkage method implementation, is done anew at each point in time prior to the construction of each new prediction. In particular, at each point in time, a model is estimated on a rolling window of size R to produce a one-step-ahead forecast. Therefore, each model under consideration uses a different subset of predictors at each point in time.

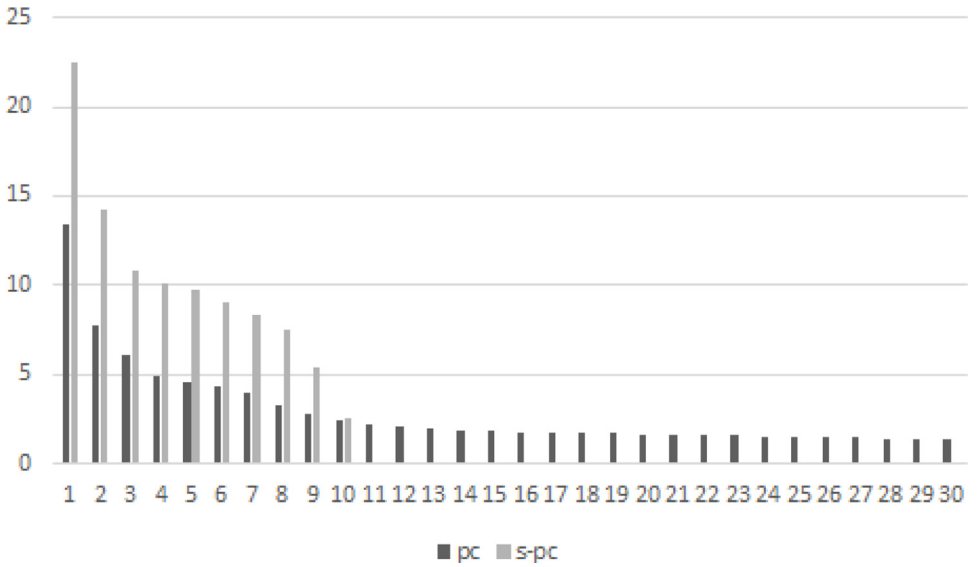


Fig. 1. Explained variance ratio for the AUD/USD currency pair.

As a benchmark model, we use a historical average (HA) model, which we calculate over the rolling window. The forecast is obtained as

$$\hat{y}_{t+1} = \frac{1}{R} \sum_{j=t-R+1}^{t} y_j. \tag{12}$$

We compare the performance of our models to the HA using an out-of-sample R^2 of Campbell and Thompson (2008):

$$R^2_{OOS} = 1 - \frac{\sum_{t=R+1}^{T}(y_t - \hat{y}_t)^2}{\sum_{t=R+1}^{T}(y_t - \bar{y}_t)^2}, \tag{13}$$

where $\hat{y}_t$ is the fitted value from the predictive regression, and $\bar{y}_t$ is the historical average return, both estimated through period $t-1$. The R^2_{OOS} is based on the mean squared errors (MSEs) of the model of interest and the benchmark model. We choose to focus on it instead of the MSE, due to the simplicity of its interpretation. In particular, the model of interest performs better than the benchmark if $R^2_{OOS} > 0$.

4. Empirical analysis

In terms of the choice of explanatory variables (predictors), we consider five different cases. We begin (Case 1) by looking at the set of all available variables. We then proceed to Case 2, where the variable set includes all information except that of the order flow. Case 3 considers measures based solely on the one-minute data.[9] In Case 4, we only use the information on a single currency pair in order to examine how much improvement we get from using cross-currency information. Case 5 includes order flow information only, but for all currency pairs at hand. We consider several data frequencies, and for each, we examine different sizes of rolling windows. For one-minute data, we consider the rolling window sizes 720 (12 h), 1440 (24 h), 7200 (five days), and 28,800 (20 days). For the hourly returns, we use 360 (15 days), 720 (30 days), 1440 (60 days), and 2880 (120 days). Note that the rolling window is moved one time step ahead, thus creating forecasts on samples that are overlapping.

In the following subsections, we first present the results for a one-hour frequency with a rolling window of 1440 (60 days) and a one-hour-ahead forecast. We then proceed to the results for one-minute data with a rolling window of 7200 (five days) and a one-minute-ahead forecasting horizon. However, prior to presenting the forecast performance results, let us focus on selected intermediate steps of the analysis. In particular, we now delve into the details of PCA and S-PCA factor estimation.

Fig. 1 presents the average amount of explained variance by each of the principal components for both the standard and the supervised approaches. The graph is for the AUD/USD currency pair and provides estimated components based on the whole set of predictors. The amount of variance explained by supervised principal components is higher than that of their non-supervised counterparts. For supervised PCA, we observe a sharp drop in the ratio after the first principal component, and then another two drops after the second and ninth components. Meanwhile, for regular PCA, there is a drop in the ratio after the first component, which then slowly declines afterwards. For the other currency pairs, one can see an even steeper decline in the amount of variance explained with each next component estimated (find the relevant graphs in Appendix B). For that reason, we restrict further analysis to the first five principal components for both approaches.

Next, let us examine the dynamics of the explained variance ratio over time. For the AUD/USD currency

[9] The motivation for this case is the ability to use the algorithm in real time and apply the same filtering—thereby excluding such variables as the average weighted spread, which requires recalculating the weights over tick-by-tick data and as a result is heavily time consuming.

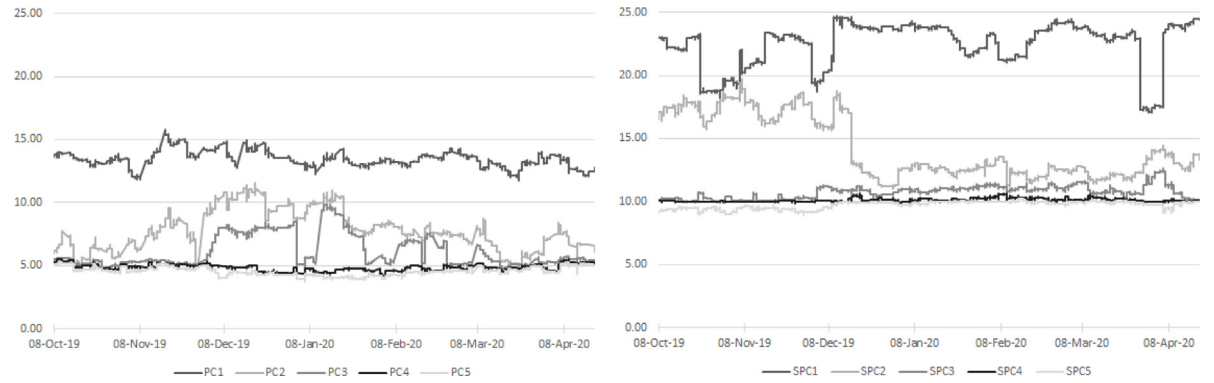


Fig. 2. Explained variance ratio over the rolling window of size 7200 (five days) for AUD/USD at a one-minute frequency.

pair,[10] we plot the explained variance ratio estimated at each point in time. Fig. 2 illustrates the dynamics for both unsupervised and supervised PCA.

It is worth noting that for some components, in the case of both unsupervised and supervised PCA, we observe drops in explained variance during several time intervals. Such patterns are noticeable for each of the currency pairs at hand. This brings us close to the developing literature on pockets of predictability, on which we elaborate at the end of this section.

4.1. Hourly returns

Table 5 presents the out-of-sample R^2 results for all cases under consideration for principal component-based models with up to five factors, as well as the two versions of LASSO and random forests.

For Case 1, which includes all predictors, we find overall very little predictability compared to the benchmark case of a regression including only a constant. For all five currency pairs, the R^2_{OOS} is negative for a model including the first PC as a predictor compared to the benchmark. Also, the model performance is generally decreasing in the number of included PCs. LASSO and random forests perform better than using PCA, but still slightly worse than a model with only a constant. Also, S-PCA struggles to outperform the benchmark, while S-PCA with one factor compares on par with the benchmark for USD/CAD, but worse for all other currency pairs.

In Case 2, we exclude the order flow from the set of predictors. For PCA, this leads to slightly worse performance for all currency pairs. For LASSO, the results are more mixed, with AUD/USD, GBP/USD, and USD/JPY performing better when order flow is excluded. For RF, all currencies apart from USD/CAD perform worse without order flow. For S-PCA, the differences from excluded order flow are very small.

In Case 3, we exclude predictors that need to be constructed from tick-level data. For PCA, this improves forecast performance for all five currency pairs for any number of PCs. Also for S-PCA, the performance is improved or unchanged but never worse. For LASSO, changes are small, with a decrease in performance for all currency pairs except USD/CAD. With random forests, the changes are also small when excluding tick-level data measures, and EUR/USD performs better but the other currencies are worse.

In Case 4, we use only predictors from the same currency pair. For S-PCA, the performance is identical to Case 1. The reason is that the supervision step excluded most cross-country variables in Case 1. PCA performs a lot better for all currencies and for any number of included PCs. For LASSO and RF, the differences are much smaller when compared to Case 1.

In Case 5, which includes only order flow, PC predicts better than in any of the other cases for all currency pairs. For S-PCA on EUR/USD with one PC, the R^2_{OOS} is -0.02%, showing predictability almost comparable to the benchmark. However, the R^2_{OOS} is still negative for all currency pairs. Also, ada-LASSO performs on par with the benchmark for USD/CAD and almost on par for AUD/USD and EUR/USD. Apart from this, the performances of LASSO and RF are similar to the other cases.

In terms of the rolling window length, we observe that the forecast results improve monotonically with the length of the rolling window for all methods.[11] This effect is especially large for PCA and S-PCA, but much smaller for RF and LASSO. LASSO in particular performs very consistently for the different rolling windows. Using an even higher rolling window size than 2880 hourly observations (120 days) may lead to improvements, but we did not pursue this, since it would make our validation sample very small.

Taken together, it seems there is very little, if any, information in cross-country variables. When these variables are included in the set of predictors, the performance of PCA deteriorates the most, since this method does not have any selection or shrinkage of variables. LASSO, S-PCA, and RF are all good at dealing with the inclusion of irrelevant predictors and therefore gain very little from having them removed a priori.

In earlier work, such as Evans and Rime (2016), it was pointed out that brokers seem to learn slowly from informed traders, making order flow a useful predictor

[10] Figures for other currency pairs, as well as for other data frequencies and rolling window sizes, are available upon request.

[11] Complete results for all rolling window sizes are unreported to conserve space but can be obtained from the authors upon request.

Table 5
Out-of-sample R^2, hourly returns.

Case 1													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−3.35	−5.99	−7.13	−7.08	−7.33	−0.11	−0.21	−0.22	−0.26	−0.31	−0.34	−0.77	−2.48
AUD/USD	−15.06	−25.86	−27.67	−27.74	−27.93	−0.29	−4.42	−5.97	−6.07	−6.18	−1.11	−0.38	−2.57
GBP/USD	−2.89	−6.04	−6.29	−6.32	−6.57	−0.54	−0.56	−0.64	−0.70	−0.77	−2.35	−4.29	−2.92
USD/JPY	−4.71	−8.11	−8.58	−8.76	−8.95	−2.61	−3.25	−3.63	−3.69	−3.70	−1.22	−0.65	−1.95
EUR/USD	−5.02	−6.97	−7.09	−7.11	−7.31	−2.42	−2.60	−2.72	−2.75	−2.73	−1.42	−0.04	−1.93
Case 2													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−4.61	−6.04	−7.11	−7.13	−7.37	−0.11	−0.17	−0.18	−0.23	−0.28	−0.33	−0.78	−2.24
AUD/USD	−19.64	−25.87	−27.68	−27.78	−27.90	−0.28	−4.45	−5.98	−6.00	−6.05	−1.04	0.75	−2.99
GBP/USD	−3.70	−5.77	−5.93	−6.10	−6.20	−0.23	−0.29	−0.36	−0.43	−0.47	−1.65	−1.58	−4.01
USD/JPY	−5.19	−8.17	−8.80	−8.89	−8.88	−2.62	−3.25	−3.63	−3.69	−3.70	−0.79	−0.95	−2.36
EUR/USD	−6.02	−6.91	−7.05	−7.07	−7.11	−2.43	−2.62	−2.71	−2.72	−2.82	−1.27	0.21	−1.57
Case 3													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−0.26	−1.81	−2.39	−2.60	−2.80	−0.11	−0.17	−0.18	−0.23	−0.28	−0.37	−0.60	−2.51
AUD/USD	−0.33	−6.43	−7.54	−8.20	−8.28	−0.34	−1.10	−1.29	−1.41	−1.42	−1.05	−0.80	−2.86
GBP/USD	−1.55	−2.98	−3.44	−3.50	−3.57	−0.23	−0.29	−0.36	−0.43	−0.47	−1.58	−1.92	−3.63
USD/JPY	−1.60	−3.89	−4.40	−4.72	−4.69	−0.46	−1.45	−1.90	−1.91	−1.93	−0.67	−0.91	−2.15
EUR/USD	−0.42	−2.43	−2.56	−2.74	−2.82	−0.13	−0.81	−1.04	−0.97	−0.99	−1.71	−0.81	−1.73
Case 4													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−0.11	−0.48	−0.73	−0.78	−0.88	−0.11	−0.21	−0.22	−0.26	−0.31	−0.25	−0.40	−2.76
AUD/USD	−0.18	−1.46	−1.42	−1.47	−1.49	−0.29	−4.42	−5.97	−6.07	−6.18	−0.46	0.25	−1.84
GBP/USD	−0.14	−0.51	−0.54	−0.60	−0.64	−0.54	−0.56	−0.64	−0.70	−0.77	−0.23	−0.75	−2.13
USD/JPY	−0.23	−1.15	−0.91	−1.13	−1.14	−2.61	−3.25	−3.63	−3.69	−3.70	−0.16	−0.36	−2.60
EUR/USD	−0.15	−0.58	−0.65	−0.70	−0.72	−2.42	−2.60	−2.72	−2.75	−2.73	−0.24	−0.34	−1.14
Case 5													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−0.08	−0.22	−0.28	−0.46	−0.62	−0.07	−0.11	−0.25	−0.41	NaN	−0.32	0.00	−2.84
AUD/USD	−0.13	−0.23	−0.53	−0.47	−0.55	−0.16	−0.36	−0.46	NaN	NaN	−0.20	−0.01	−2.12
GBP/USD	−0.22	−0.32	−0.27	−0.42	−0.49	−0.26	−0.37	−0.31	−0.47	−0.54	−0.30	−0.06	−3.19
USD/JPY	−0.15	−0.38	−0.44	−0.56	−0.64	−0.27	NaN	NaN	NaN	NaN	−0.24	−0.06	−2.18
EUR/USD	−0.09	−0.25	−0.30	−0.46	−0.51	−0.02	−0.16	NaN	NaN	NaN	−0.35	−0.03	−2.33

Note: Shows out-of-sample R^2 for hourly returns for PCA with one to five components (pc1–pc5), supervised PCA (s-pc1–s-pc5), LASSO, adaptive LASSO (ada-LAsso), and random forests (RF) across five predictor subsets. Case 1 uses all predictors; Case 2 omits order flow; Case 3 omits tick-level measures; Case 4 uses only same-currency predictors; and Case 5 uses only order flow. Rolling window size: 1440 (60 days).

for a relatively long time such as one hour or even days. This no longer seems to be the case on modern limit order book exchanges such as LMAX, where we find very little predictability one hour ahead from any information in the order book. We therefore continue to look for predictability at the one-minute horizon.

4.2. One-minute returns

For the one-minute data, the forecast errors improve with the rolling window size but only up to 7200 (five days) observations. For 28,800 (20 days) observations, the results are almost identical to 7200. Therefore, we present the results in Table 6 for a rolling window of 7200 observations.

For Case 1, the predictability one minute ahead is generally higher than one hour ahead with the largest improvement for PCA and S-PCA. The LASSO and RF perform roughly the same as at the hourly frequency. However, also for one minute ahead, all models for all currency pairs perform worse out of sample than the benchmark that only includes a constant. Removing order flow (Case 2) leads to very minor differences in the results for all methods and currency pairs. For PCA, removing all predictors that require tick data (Case 3) increases performance somewhat for all currency pairs. For LASSO, S-PCA, and RF, the performance is very similar to Case 1 with all included predictors. Comparing with one-hour forecasts, the performance is generally better in Case 3 for the one-minute frequency, especially for PCA. When looking at information from a single currency pair (Case 4), the pattern is the same as for one-hour forecasts: PCA performs better, while S-PCA performs exactly as in Case 1. The LASSO forecasts are better for USD/CAD, AUD/USD, and EUR/USD. For EUR/USD, the LASSO also performs better than the benchmark model, but with a very slim margin. The RF performs worse for all currency pairs when cross-currency information is excluded. When only order flow is used as a predictor (Case 5), the performance is very similar to the one-hour frequency for PCA, S-PCA, and LASSO; RF, however, performs worse.

Taken together, the predictability at the one-minute and one-hour forecast horizons seems to be about the same. For PCA, S-PCA, and LASSO, the performance is very similar, but for random forests, the performance is worse at the one-minute frequency. We also looked at five-minute, 15-minute, two-hour, six-hour, and daily forecast

Table 6
Out-of-sample R^2, one-minute returns.

Case 1													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−0.46	−0.59	−0.77	−0.92	−1.04	−0.04	−0.08	−0.09	−0.11	−0.11	−0.30	−1.36	−1.16
AUD/USD	−0.83	−0.90	−1.03	−1.12	−1.24	−0.06	−0.10	−0.12	−0.14	−0.15	−1.18	−1.87	−1.42
GBP/USD	−0.93	−1.09	−1.26	−1.42	−1.55	−0.03	−0.07	−0.11	−0.18	−0.27	−0.42	−1.37	−1.06
USD/JPY	−0.85	−0.89	−1.02	−1.10	−1.29	−0.14	−0.24	−0.29	−0.33	−0.33	−0.40	−0.71	−1.60
EUR/USD	−0.49	−0.69	−0.78	−0.88	−1.01	−0.14	−0.32	−0.37	−0.39	−0.42	−0.17	−0.31	−1.15
Case 2													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−0.46	−0.59	−0.75	−0.87	−0.99	−0.02	−0.03	−0.04	−0.05	−0.05	−0.35	−1.38	−1.22
AUD/USD	−0.81	−0.88	−1.00	−1.10	−1.22	−0.06	−0.14	−0.36	−0.41	−0.46	−1.10	−1.33	−1.63
GBP/USD	−0.97	−1.15	−1.30	−1.46	−1.56	−0.03	−0.06	−0.10	−0.18	−0.23	−0.51	−1.35	−0.96
USD/JPY	−0.82	−0.85	−0.97	−1.04	−1.24	−0.14	−0.25	−0.25	−0.26	−0.28	−0.34	−0.70	−1.52
EUR/USD	−0.49	−0.69	−0.76	−0.85	−0.97	−0.14	−0.29	−0.34	−0.37	−0.38	−0.23	−0.32	−1.17
Case 3													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−0.21	−0.34	−0.48	−0.60	−0.69	−0.02	−0.03	−0.04	−0.05	−0.05	−0.33	−1.29	−1.26
AUD/USD	−0.25	−0.40	−0.48	−0.58	−0.68	−0.05	−0.07	−0.11	−0.18	−0.40	−1.02	−1.11	−1.73
GBP/USD	−0.34	−0.49	−0.61	−0.80	−0.88	−0.03	−0.06	−0.10	−0.18	−0.23	−0.50	−1.21	−1.33
USD/JPY	−0.32	−0.41	−0.51	−0.60	−0.77	−0.14	−0.25	−0.25	−0.26	−0.28	−0.28	−0.64	−1.68
EUR/USD	−0.32	−0.49	−0.56	−0.65	−0.78	−0.15	−0.27	−0.33	−0.35	−0.35	−0.21	−0.36	−1.28
Case 4													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−0.15	−0.21	−0.26	−0.27	−0.28	−0.04	−0.08	−0.09	−0.11	−0.11	−0.07	−0.31	−2.42
AUD/USD	−0.13	−0.17	−0.19	−0.20	−0.23	−0.06	−0.10	−0.12	−0.14	−0.15	−0.14	−0.46	−2.36
GBP/USD	−0.13	−0.18	−0.22	−0.23	−0.26	−0.03	−0.07	−0.11	−0.18	−0.27	−0.19	−0.20	−1.77
USD/JPY	−0.18	−0.27	−0.38	−0.42	−0.43	−0.14	−0.24	−0.29	−0.33	−0.33	−0.19	−0.14	−2.57
EUR/USD	−0.24	−0.29	−0.30	−0.32	−0.32	−0.14	−0.32	−0.37	−0.39	−0.42	−0.02	−0.15	−2.06
Case 5													
Currency	**pc1**	**pc2**	**pc3**	**pc4**	**pc5**	**s-pc1**	**s-pc2**	**s-pc3**	**s-pc4**	**s-pc5**	**LASSO**	**ada-LASSO**	**RF**
USD/CAD	−0.07	−0.04	−0.01	0.02	0.00	−0.07	−0.04	−0.01	0.02	0.00	0.02	−0.01	−3.25
AUD/USD	−0.06	−0.06	−0.07	−0.08	−0.08	−0.04	−0.06	−0.07	−0.08	NaN	−0.08	−0.01	−2.69
GBP/USD	−0.14	−0.13	−0.09	−0.10	−0.07	−0.01	−0.02	NaN	NaN	NaN	0.02	−0.01	−2.93
USD/JPY	−0.09	−0.11	−0.11	−0.13	−0.12	−0.03	−0.06	NaN	NaN	NaN	−0.09	0.00	−3.15
EUR/USD	−0.04	−0.05	−0.04	−0.04	0.01	0.00	−0.02	−0.02	0.01	NaN	0.02	−0.01	−2.71

Note: Shows out-of-sample R^2 for one-minute returns for PCA with one to five components (pc1–pc5), supervised PCA (s-pc1–s-pc5), LASSO, adaptive LASSO (ada-LAsso), and random forests (RF) across five predictor subsets. Case 1 uses all predictors; Case 2 omits order flow; Case 3 omits tick-level measures; Case 4 uses only same-currency predictors; and Case 5 uses only order flow. Rolling window size: 7200 (five days).

frequencies. The results were qualitatively similar and available upon request. Overall, it does not seem possible to predict FX returns at any frequency ranging from one minute to one day using limit order book information.

4.3. Pockets of predictability

Recently, Farmer et al. (2023) found that short-term predictability for equity returns varies significantly over time, with some periods exhibiting pockets of predictability. In addition, they found substantial variation over time regarding which variables have predictive power. To further investigate this for high-frequency FX returns, we take a closer look at which variables are selected by the (adaptive) LASSO. We display results for Case 1 at the one-minute frequency in Table 7.

For the USD/CAD exchange rate, the predictor that is included most often is order flow for the same currency. This is true for both LASSO and adaptive LASSO. The second and third most often included variables are cross-currency information on order flow from EUR/USD and AUD/USD. Also, for AUD/USD, GBP/USD, and USD/JPY, order flow from the same currency is the predictor most often included. The only exception is the EUR/USD exchange rate, for which the order flow of USD/CAD is the most often included predictor, and order flow from the same currency is the second most often included predictor. The only variables that are among the three most selected for any of the currency pairs are order book imbalances (IMB2 and IMB3) and then always for the same currency. Overall, the results for LASSO and adaptive LASSO are very similar, with the same predictors always being the most common. Taken together, the results show order flow to be the most important predictor and same-currency information to be more important than cross-currency information. We now turn to look at how predictor selection varies over time.

Figs. 3 and 4 show when the 10 most commonly selected predictors are included using non-overlapping samples[12] for the AUD/USD and EUR/USD currency pairs, respectively. The top row shows the out-of-sample R^2 for

[12] Plotting all our rolling window estimation results would create artificial dependence over time, so we plot non-overlapping samples by plotting every 7200th estimate, since our rolling window size is 7200 observations.

Table 7
LASSO variable selection.

Currencies	LASSO	% included	ada-LASSO	% included
USD/CAD	OrdFlow-USDCAD	62.11	OrdFlow-USDCAD	62.86
USD/CAD	OrdFlow-EURUSD	56.46	OrdFlow-EURUSD	56.46
USD/CAD	OrdFlow-AUDUSD	49.03	OrdFlow-AUDUSD	49.03
AUD/USD	IMB3-AUDUSD	51.26	IMB3-AUDUSD	51.26
AUD/USD	IMB2-AUDUSD	46.06	IMB2-AUDUSD	46.06
AUD/USD	OrdFlow-USDCAD	44.57	OrdFlow-EURUSD	44.57
GBP/USD	OrdFlow-GBPUSD	52.00	OrdFlow-GBPUSD	51.26
GBP/USD	OrdFlow-EURUSD	45.32	OrdFlow-EURUSD	46.06
GBP/USD	OrdFlow-USDCAD	41.60	OrdFlow-USDCAD	42.34
USD/JPY	OrdFlow-USDJPY	53.20	OrdFlow-USDJPY	53.20
USD/JPY	OrdFlow-AUDUSD	46.51	OrdFlow-AUDUSD	46.51
USD/JPY	OrdFlow-EURUSD	42.34	OrdFlow-EURUSD	40.86
EUR/USD	OrdFlow-USDCAD	58.69	OrdFlow-USDCAD	58.69
EUR/USD	OrdFlow-EURUSD	54.23	OrdFlow-EURUSD	54.23
EUR/USD	IMB3-EURUSD	39.37	IMB3-EURUSD	41.60

Note: This table shows the three variables that are most often kept for each of the five currency pairs in the LASSO and adaptive LASSO regressions for Case 1 at the one-minute frequency. The percentage included is the fraction of times the variable is kept in the LASSO regression over the total number of 193,837 estimation windows. The results are for rolling window estimates with a window size of 7200 min (five days).

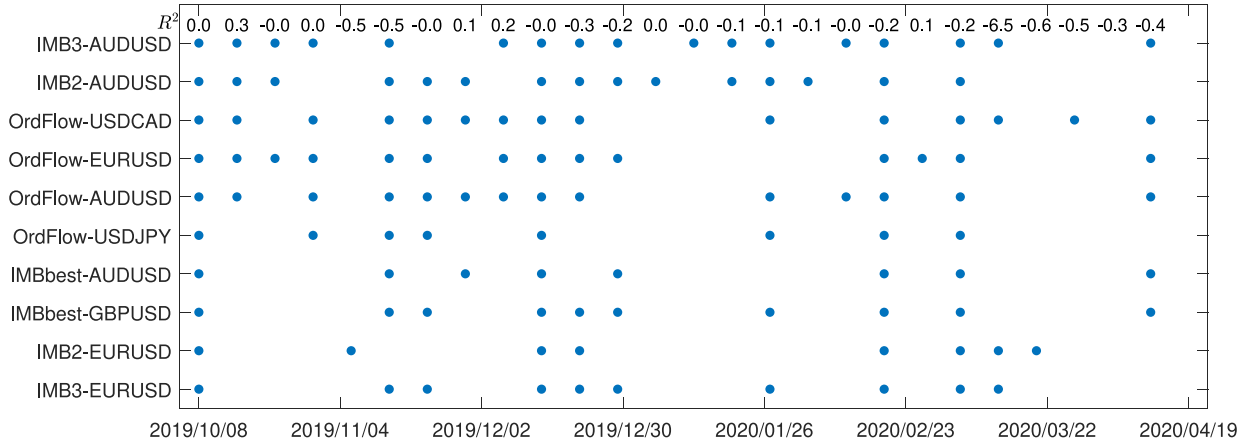


Fig. 3. Based on AUD/USD LASSO one-minute data, out-of-sample R^2 (top row) for the inclusion of the 10 most frequently selected predictors using non-overlapping windows over the data span (bottom row).

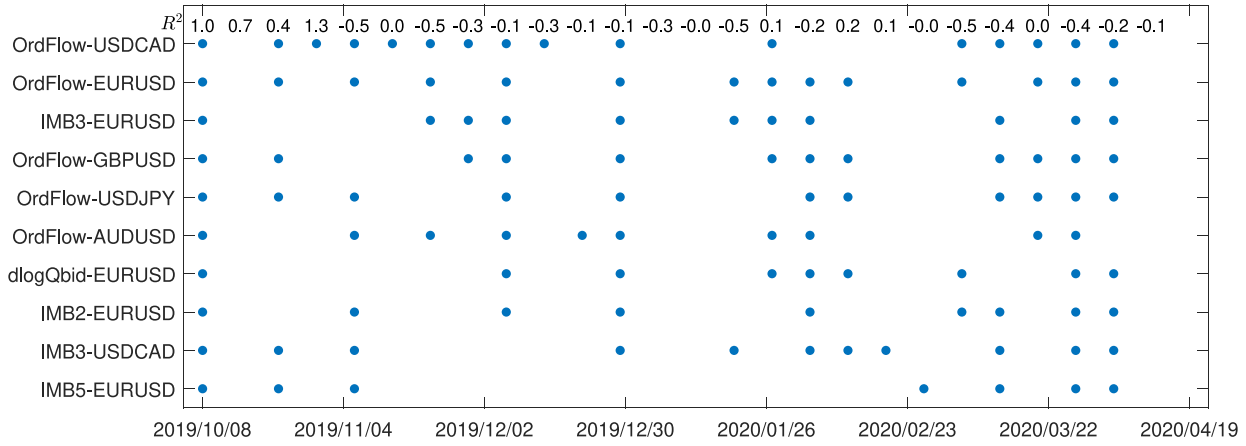


Fig. 4. Based on EUR/USD LASSO one-minute data, out-of-sample R^2 (top row) for the inclusion of the 10 most frequently selected predictors using non-overlapping windows over the data span (bottom row).

the corresponding time period. For AUD/USD, the order book imbalance measures (IMB2 and IMB3) are more often included towards the beginning of the sample than towards the end. For the cross-currency predictors IMB2 and IMB3 from EUR/USD, the pattern is the opposite, with these predictors being included more often towards the end. Overall, time variation seems important, with no predictor being consistently selected over the whole sample. Further, there is much variation over time in the number of included predictors, indicating that pockets of predictability also exist in high-frequency FX markets. However, there is no clear pattern between the included predictors and the predictive ability of the model.

5. Conclusions

This paper contributes to the field of foreign exchange rate forecasting by exploring the potential of detailed limit order book information and considering cross-currency pair information. The study addressed the limited literature on the predictability of exchange rates in comparison to equity markets, despite the immense size of the FX market. It shed light on the lack of significance of microstructure information and its implications for short-term forecast accuracy.

Furthermore, our research emphasized the importance of employing dimension reduction techniques to handle the abundance of predictors effectively. In particular, we examined the application of factor-augmented regression techniques, including PCA and supervised PCA, to efficiently handle the abundance of predictors obtained from the LOB data. Additionally, we incorporated the LASSO and random forest methods as competing models.

Throughout the analysis, we used a rolling window approach to accommodate potential changes in market behavior and obtain out-of-sample forecasts at various frequencies. The investigation spanned different cases for the initial set of predictors, enabling a comprehensive evaluation of the impact of cross-currency information and the inclusion of irrelevant predictors. Our findings revealed that despite rich LOB information, the overall predictability of exchange rates is low across models and time frames, reinforcing the efficient market hypothesis in the context of forex markets. Cross-currency variables do not notably improve predictive accuracy. However, the findings of some predictability in relation to the order flow variable suggest the potential existence of transient market inefficiencies and the gradual assimilation of microstructural information into prices.

For future work, it would be interesting to go beyond statistical evaluation metrics and to explore the potential profitability of a trading strategy as suggested by Melvin et al. (2013).

CRediT authorship contribution statement

Yana Petrova: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Anders Vilhelmsson:** Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lars L. Nordén:** Supervision, Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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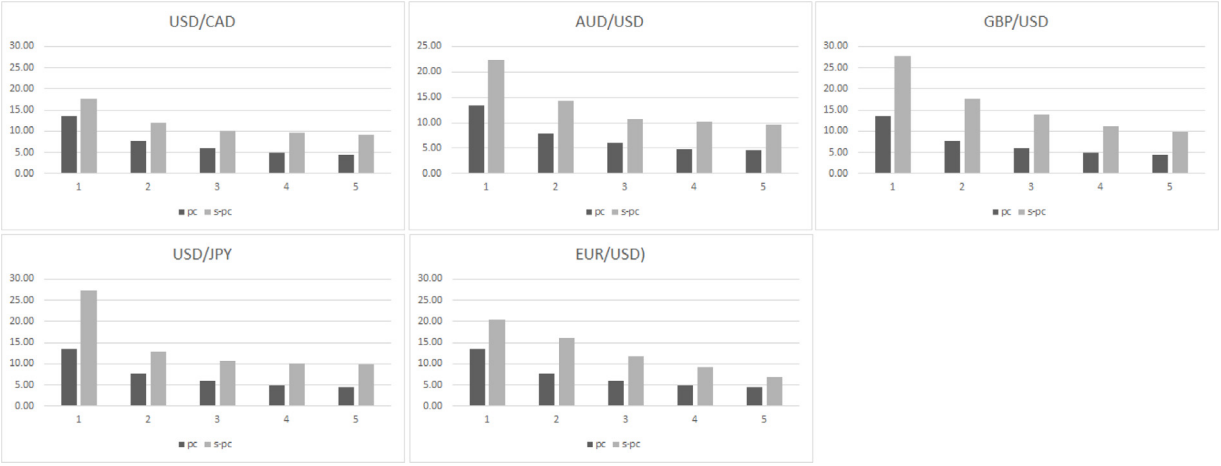


Table B.8
Out-of-Sample R^2, hourly returns.

Case 1

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−2.72	−2.85	−4.61	−6.13	−5.83	−1.72	−0.52	−4.67	−8.75	−11.80	0.50	−3.43
AUD/USD	−1.25	−1.26	−1.47	−1.84	4.10	−0.54	0.53	2.19	4.66	6.22	−0.36	17.30
GBP/USD	−5.75	−11.51	−5.47	−8.77	−13.65	−3.18	−13.07	−9.77	−12.35	−17.29	−15.11	−7.58
USD/JPY	−9.54	−15.43	−21.14	−26.63	−38.27	−6.58	−19.90	−26.46	−76.91	−107.48	−86.90	−18.45
EUR/USD	−12.75	−13.82	−9.42	−12.48	−18.69	−10.26	−17.62	−16.58	−18.41	−15.44	−16.07	−1.02

Case 2

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−2.53	−2.48	−4.44	−5.86	−6.10	−1.72	−0.69	−4.81	−7.34	−10.60	0.48	−3.47
AUD/USD	−1.20	−1.33	−1.07	−1.25	4.01	−0.48	−0.37	2.55	4.86	8.25	−0.36	17.11
GBP/USD	−6.16	−11.88	−8.20	−9.88	−14.19	−3.38	−12.83	−9.49	−12.47	−14.58	−23.57	−5.70
USD/JPY	−8.20	−13.46	−19.48	−26.55	−38.61	−6.00	−18.93	−23.72	−70.35	−105.12	−80.41	−20.65
EUR/USD	−13.60	−15.17	−13.63	−13.32	−18.98	−10.88	−18.24	−18.54	−19.55	−15.89	−12.77	−2.06

Case 3

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−1.70	−1.34	−3.26	−4.78	−7.45	−0.95	−2.06	−3.44	−9.70	−10.37	−2.46	−4.64
AUD/USD	−1.63	−1.39	0.54	−0.07	4.32	−0.24	1.56	4.87	5.60	10.30	−0.36	13.17
GBP/USD	−3.55	−6.33	−5.93	−4.02	−7.67	−2.84	−8.21	−9.09	−9.35	−8.73	−12.89	−4.37
USD/JPY	−6.56	−19.41	−17.97	−20.50	−42.86	−7.13	−19.83	−67.75	−73.00	−81.08	−42.29	−23.45
EUR/USD	−6.69	−7.22	−5.68	−6.20	−5.90	−7.06	−7.97	−8.45	−10.17	−8.66	−13.38	−1.72

Case 4

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.03	−2.32	−5.80	−5.93	−4.54	−1.72	−0.52	−4.67	−8.75	−11.80	1.06	−7.71
AUD/USD	−0.13	−0.78	−0.78	0.04	1.30	−0.54	0.53	2.19	4.66	6.22	2.55	19.39
GBP/USD	−9.22	−8.46	−9.81	−11.71	−11.30	−3.18	−13.07	−9.77	−12.35	−17.29	−6.85	−6.66
USD/JPY	−5.64	−6.92	−7.26	−8.71	−8.66	−6.58	−19.90	−26.46	−76.91	−107.48	1.05	−18.54
EUR/USD	2.80	0.44	−0.15	−0.55	−0.79	−10.26	−17.62	−16.58	−18.41	−15.44	−1.21	−7.77

Case 5

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.47	−0.71	−0.18	−0.54	−0.18	−0.96	−1.20	−0.66	−1.02	−0.67	0.19	−4.97
AUD/USD	−0.66	−0.25	−0.42	−0.50	−0.54	−0.62	−0.61	0.00	0.00	0.00	−0.39	6.51
GBP/USD	−0.63	−1.26	−2.25	−3.84	−3.80	−1.60	−4.19	−4.28	0.00	0.00	−5.49	−14.16
USD/JPY	−1.01	−0.28	−1.43	−3.87	−3.84	0.74	−3.78	0.00	0.00	0.00	0.47	−30.26
EUR/USD	−0.26	−0.64	−1.35	−1.59	−1.93	−1.19	−1.58	−2.29	−2.54	−2.88	−0.43	−4.80

Note: Shows out-of-sample R^2 for hourly returns for PCA with one to five components (pc1—pc5), supervised PCA (s-pc1–s-pc5), LASSO, and random forests (RF) across five predictor subsets. Case 1 uses all predictors; Case 2 omits order flow; Case 3 omits tick-level measures; Case 4 uses only same-currency predictors; and Case 5 uses only order flow. Rolling window size: 1440 (60 days).

data are proprietary and cannot be shared, but the measures constructed from the data used for the main analysis can be shared at an aggregate level.

Appendix A. PC explained variance ratio

Appendix B. Replication of main results on the EBS platform

Since our main analysis is based on LMAX data, we also want to check whether our results hold on an exchange with a higher market share and, therefore, a potentially more informative order book. We use data from the Electronic Broking Services (EBS) during January 2015,[13] which is a limit order book operated by CME Group. While it is hard to find exact market share information, EBS is the leading platform for USD/JPY and EUR/USD, whereas Reuters handles the majority of trades in AUD/USD, CAD/USD, and GBP/USD; see for example Ito et al. (2012). The same dataset is used by Hagströmer and Menkveld (2019), who provide further details on the data.

If the relatively low market share of LMAX were the reason we did not find predictability in the main analysis, we would expect to see increased predictability for USD/JPY and EUR/USD, where EBS has the majority of traded volume. This is not the case. Instead, we find no predictability for any of the frequencies, predictor cases, or estimation methods (see Tables B.8–B.10). It is therefore unlikely that our main results are driven by LMAX's relatively low market share. We also examine a 30-second frequency without any increase in predictability. The only case in which we find predictability is for the AUD/USD currency pair, estimated using random forests and supervised PCA at the one-hour frequency. Here, we observe

[13] We use this specific time period because the data were available to us. Furthermore, market efficiency is generally assumed to increase over time, so using an older sample than in our main analysis should increase the chance of finding predictability.

Table B.9
Out-of-Sample R^2, one-minute returns.

Case 1

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−1.76	−1.83	−2.13	−2.32	−2.67	−0.26	−0.57	−0.72	−0.95	−1.47	0.19	−0.26
AUD/USD	−1.62	−1.64	−1.80	−2.03	−2.19	−0.23	−0.52	−0.62	−0.75	−0.80	−0.10	−0.97
GBP/USD	−2.64	−2.83	−3.39	−3.34	−3.43	−1.96	−2.02	−2.12	−2.32	−2.41	−1.11	−1.49
USD/JPY	−4.56	−4.64	−4.93	−5.31	−5.93	−0.38	−0.91	−1.66	−1.71	−1.77	−7.42	−5.17
EUR/USD	−1.22	−1.46	−1.46	−1.52	−1.55	−0.20	−0.93	−1.15	−1.21	−1.31	−1.40	−1.93

Case 2

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.01	0.00	−0.03	−0.04	−0.04	0.00	0.02	0.01	0.00	*(NA)*	−0.01	−0.61
AUD/USD	−0.01	−0.01	−0.02	−0.03	−0.04	−0.02	−0.02	0.00	0.00	*(NA)*	−0.02	−0.93
GBP/USD	0.01	0.01	−0.01	−0.04	−0.05	−0.01	0.04	0.00	0.00	*(NA)*	−0.10	−1.74
USD/JPY	−0.01	−0.02	−0.03	−0.06	−0.08	−0.06	−0.03	−0.05	−0.06	*(NA)*	−0.07	−1.74
EUR/USD	0.00	−0.02	−0.03	−0.06	−0.07	−0.05	0.01	−0.05	−0.05	*(NA)*	−0.12	−1.56

Case 3

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−1.76	−1.82	−2.12	−2.33	−2.67	−0.26	−0.58	−0.72	−0.95	−1.47	0.20	−0.42
AUD/USD	−1.62	−1.64	−1.81	−2.04	−2.19	−0.23	−0.53	−0.69	−0.73	−0.90	−0.10	−1.11
GBP/USD	−2.64	−2.82	−3.37	−3.33	−3.42	−1.96	−2.02	−2.12	−2.31	−2.35	−1.09	−1.67
USD/JPY	−4.55	−4.64	−4.93	−5.30	−5.94	−0.38	−0.91	−1.66	−1.71	−1.77	−7.28	−4.59
EUR/USD	−1.22	−1.46	−1.46	−1.51	−1.54	−0.20	−0.92	−1.15	−1.21	−1.34	−0.94	−1.98

Case 4

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.37	−0.66	−0.89	−1.05	−1.24	−0.26	−0.58	−0.72	−0.95	−1.47	−0.25	−0.40
AUD/USD	−0.44	−0.49	−0.59	−0.80	−0.89	−0.06	−0.33	−0.55	−0.84	−0.93	−0.65	0.21
GBP/USD	−0.34	−0.70	−1.10	−1.54	−1.64	−0.06	−0.14	−0.36	−0.41	−0.41	−0.99	−1.88
USD/JPY	−1.21	−2.46	−2.79	−3.14	−3.63	−0.24	−0.49	−0.77	−0.95	−0.93	−3.45	−3.62
EUR/USD	−0.43	−0.53	−0.56	−0.73	−0.71	−0.18	−0.11	−0.15	−0.15	−0.18	−0.48	−1.26

Case 5

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.26	−1.93	−1.89	−1.79	−2.18	−0.26	−0.57	−0.72	−0.95	−1.47	−0.26	−0.20
AUD/USD	0.00	−1.97	−2.02	−1.88	−1.83	−0.23	−0.52	−0.62	−0.75	−0.80	−0.05	−1.18
GBP/USD	−0.91	−1.38	−1.62	−1.76	−1.73	−1.96	−2.02	−2.12	−2.32	−2.41	−0.41	−2.99
USD/JPY	−0.29	−0.47	−0.51	−0.53	−0.58	−0.38	−0.91	−1.66	−1.71	−1.77	−0.31	−3.88
EUR/USD	−0.26	−0.34	−0.33	−0.37	−0.41	−0.20	−0.93	−1.15	−1.21	−1.31	0.10	−2.55

Note: Shows out-of-sample R^2 for one-minute returns for PCA with one to five components (pc1—pc5), supervised PCA (s-pc1–s-pc5), LASSO, and random forests (RF) across five predictor subsets. Case 1 uses all predictors; Case 2 omits order flow; Case 3 omits tick-level measures; Case 4 uses only same-currency predictors; and Case 5 uses only order flow. Rolling window size: 7200 (five days).

Table B.10
Out-of-sample R^2, 30-second returns.

Case 1

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.42	−0.60	−0.89	−1.11	−1.31	−0.12	−0.10	−0.08	−0.72	−1.23	−1.88	−17.75
AUD/USD	−0.90	−1.73	−2.27	−2.58	−2.81	−0.43	−0.87	−0.94	−1.25	−1.29	−26.80	−147.54
GBP/USD	−0.54	−1.22	−1.30	−2.17	−2.56	−0.52	−0.89	−1.36	−1.72	−2.01	−36.68	−6.37
USD/JPY	−1.35	−2.29	−2.63	−3.13	−3.48	−0.53	−0.70	−1.34	−2.03	−2.12	−4.95	−96.64
EUR/USD	−0.56	−1.10	−1.18	−1.60	−1.71	−0.13	−0.13	−0.19	−0.23	−0.22	−0.86	−0.11

Case 2

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.42	−0.60	−0.90	−1.13	−1.33	−0.12	−0.10	−0.08	−0.88	−1.25	−1.94	−10.81
AUD/USD	−0.90	−1.73	−2.28	−2.58	−2.80	−0.42	−0.87	−0.94	−1.29	−1.31	−26.65	−161.67
GBP/USD	−0.54	−1.21	−1.30	−2.18	−2.55	−0.52	−0.89	−1.36	−1.71	−2.04	−36.43	−6.66
USD/JPY	−1.34	−2.29	−2.62	−3.14	−3.48	−0.53	−0.69	−1.34	−2.03	−2.21	−4.96	−93.24
EUR/USD	−0.56	−1.10	−1.18	−1.60	−1.71	−0.13	−0.13	−0.19	−0.23	−0.21	−0.86	−0.12

Case 3

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.25	−0.41	−0.73	−1.01	−1.02	−0.09	−0.10	−0.14	−0.90	−0.71	−1.90	−10.62
AUD/USD	−0.83	−1.48	−2.01	−2.49	−2.53	−0.42	−0.89	−0.97	−1.14	−1.22	−27.46	−183.17
GBP/USD	−0.38	−0.80	−0.96	−1.31	−1.37	−0.28	−0.56	−0.75	−1.20	−1.44	−33.96	−8.30
USD/JPY	−1.27	−1.76	−2.30	−2.59	−2.90	−0.23	−0.51	−1.64	−2.34	−2.60	−5.24	−103.07
EUR/USD	−0.55	−1.10	−1.11	−1.35	−1.58	−0.14	−0.14	−0.18	−0.22	−0.34	−0.81	−0.15

Case 4

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.27	−0.59	−0.50	−0.69	−0.80	−0.12	−0.10	−0.08	−0.72	−1.23	−1.14	−0.84
AUD/USD	−0.31	−0.86	−0.97	−1.07	−1.53	−0.43	−0.87	−0.94	−1.25	−1.29	−6.31	−158.09
GBP/USD	−0.67	−1.38	−2.00	−2.33	−2.42	−0.52	−0.89	−1.36	−1.72	−2.01	−30.86	−7.12
USD/JPY	−0.11	−0.73	−0.93	−0.95	−0.99	−0.53	−0.70	−1.34	−2.03	−2.12	−0.61	−44.39
EUR/USD	−0.09	−0.16	−0.22	−0.25	−0.26	−0.13	−0.13	−0.19	−0.23	−0.22	−0.57	−0.29

Case 5

Currency	pc1	pc2	pc3	pc4	pc5	s-pc1	s-pc2	s-pc3	s-pc4	s-pc5	LASSO	RF
USD/CAD	−0.02	−0.01	−0.04	−0.04	−0.04	−0.15	−0.02	−0.03	−0.04	−0.04	0.00	−0.58
AUD/USD	−0.03	−0.04	−0.05	−0.06	−0.07	−0.15	−0.02	−0.03	−0.04	−0.04	−0.01	−0.78
GBP/USD	−0.04	−0.08	−0.10	−0.11	−0.13	−0.02	0.00	−0.03	−0.04	0.00	−0.04	−1.15
USD/JPY	−0.02	−0.03	−0.06	−0.06	−0.07	−0.40	−0.02	−0.02	−0.04	0.00	−0.03	−0.56
EUR/USD	0.00	0.00	0.00	0.00	0.00	−0.02	−0.02	0.00	0.00	0.00	0.00	−0.72

Note: Shows out-of-sample R^2 for 30-second returns for PCA with one to five components (pc1—pc5), supervised PCA (s-pc1–s-pc5), LASSO, and random forests (RF) across five predictor subsets. Case 1 uses all predictors; Case 2 omits order flow; Case 3 omits tick-level measures; Case 4 uses only same-currency predictors; and Case 5 uses only order flow. Rolling window size: 7200 (five days).

R^2_{OOS} values of up to 19.4% for Case 4 with random forests and up to 10.3% with supervised PCA. We therefore find it unlikely that our main results are due to either LMAX's relatively low market share or our use of one-minute returns as the highest frequency in the main analysis.

Data and code availability

All data and code necessary to reproduce the work presented in this manuscript are available at https://github.com/yanapetrova/FX_Order-Book.

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